3.4 Hypothesis testing

Vladimir Buskin

English Language and Linguistics, Catholic University of Eichstätt-Ingolstadt

ORCID: 0009-0005-5824-1012

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This handout introduces the principles of scientific inference and null hypothesis significance testing (NHST) for both categorical and continuous linguistic data, covering hypothesis formulation, test statistics, sampling distributions, *p*-values, Type I/II errors, statistical power, and common misinterpretations of significance, with illustrative examples from linguistic research.

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Suggested reading

For linguists:

Gries (2021): Chapter 1.3.2

General:

Baguley (2012): Chapter 4

Agresti and Kateri (2022): Chapter 5

Dienes (2008)

On scientific inference

Science begins and ends with theory, and statistics acts as the "go-between". Regardless of the discipline, solid research is characterised by a robust theoretical foundation that gives rise to **substantive hypotheses**, i.e., theory-driven predictions about a population of interest. From this rather concrete hypothesis, it should be possible to derive a **statistical hypothesis** that re-states the prediction in more formal/mathematical terms. After checking it against real-world data, researchers can either **confirm** or **reject** their hypothesis, after which they may decide to amend (or even abandon) their theory – or keep it as is.

Null hypothesis significance testing (NHST)

The **NHST** framework offers researchers a convenient way of testing their theoretical assumptions. This chiefly involves setting up a set of (ideally) falsifiable statistical hypotheses, gathering evidence from the observed data and computing the (in)famous 'p-value' to determine "statistical significance" – a notion that is frequently misinterpreted in scientific studies.

> **i** Is this the only way of testing hypotheses?
>
> The answer is a resounding **no**. Despite its immense popularity, NHST is problematic in many respects and hence subject to heavy criticism (cf. Dienes (2008): 76; Baguley (2012): 143-144). There are other statistical schools that can remedy many of its shortcomings and come with distinct advantages, such as those relying on likelihood-based inference and Bayesian principles. Although these are also becoming increasingly common in linguistics, they are still restricted to very few sub-disciplines and journals (mostly in the area of psycholinguistics).

H_0 vs. H_1

Statistical hypotheses always come in pairs: A **null hypothesis** is accompanied by an **alternative hypothesis**. They are set up before (!) seeing the data and justified by previous research. Note that these hypotheses "always make a statement about the population" (Casella and Berger 2002: 373).

- The **null hypothesis** H_0 describes the "default state of the world" (James et al. 2021: 555). It claims there is no noteworthy effect to be observed in the data.

- The **alternative hypothesis** H_1 (or H_a) plainly states that H_0 is false, suggesting that there is an effect of some kind.

Categorical data

We are interested in finding out whether the `Type` of an English genitive ('s' vs. 'of') depends on `Possessor Animacy` ('animate' vs. 'inanimate'). Our hypotheses are:

- H_0 : Genitive `Type` and `Possessor Animacy` are independent.
- H_1 : Genitive `Type` and `Possessor Animacy` are **not** independent.

  According to a χ^2-test of independence, there is a statistically significant relationship between genitive type and possessor animacy ($p < 0.001$, $\chi^2 = 106.44$, $df = 1$).

i What does independence really mean?

The core idea is "that the probability distribution of the response variable is the same for each group" (Agresti and Kateri 2022: 177). Assume genitive `Type` is the dependent variable and `Possessor Animacy` the independent variable. Then independence would entail that the probabilities of the outcomes of the dependent variable `Type = "s"` and `Type = "of"` are not influenced by whether they occur in the groups `Possessor Animacy = "animate"` or `Possessor Animacy = "inanimate"`.
If we consider two variables at the same time, such as X and Y, they are said to have **marginal probability functions**, which we can call $f_1(\text{Type})$ and $f_2(\text{Animacy})$ here. If we condition the outcomes of both variables on each other, the following equivalence will hold if the variables are independent:

$$f(\text{Type} \mid \text{Animacy}) = f_1(\text{Type}) \text{ and } f(\text{Animacy} \mid \text{Type}) = f_2(\text{Animacy}). \quad (1)$$

Thus, the null hypothesis assumes that the probabilities of each combination of values (such as `Type` and `Possessor Animacy`), denoted by π_{ij}, have the relationship in Equation 1. This can be stated succinctly as

$$H_0 : \pi_{ij} = P(X = j)P(Y = i), \quad (2)$$

where $P(X = j)$ denotes the the probability that the random variable X takes the value j and $P(Y = i)$ the probability that the random variable Y takes the value i.
For illustration, consider the data below:

		Possessor Animacy	
		animate	inanimate
Genitive	s	π_{11}	π_{12}
	of	π_{21}	π_{22}

Independence would be given if for each cell the following relationships would hold:

$$\pi_{11} = P(X = \text{animate})P(Y = \text{s}) \tag{3}$$
$$\pi_{12} = P(X = \text{inanimate})P(Y = \text{s}) \tag{4}$$
$$\pi_{21} = P(X = \text{animate})P(Y = \text{of}) \tag{5}$$
$$\pi_{22} = P(X = \text{inanimate})P(Y = \text{of}) \tag{6}$$
$$\tag{7}$$

Continuous data

As part of a phonetic study, we compare the base frequencies of the F1 formants of vowels (in Hz) for male and female speakers of Apache. We forward the following hypotheses:

- H_0 : mean `F1 frequency` of men = mean `F1 frequency` of women.
- H_1 : mean `F1 frequency` of men $\neq$ mean `F1 frequency` of women.

  According to a two-sample t-test, there is a significant difference between the mean F1 frequencies of male and female speakers of Apache ($t(112.19) = 2.44$, $p < 0.05$).

Test statistics

To facilitate the decision-making process, we proceed to gather statistical evidence from the observed data, that is, the sample. Since NHST primarily revolves around H_0 (and not H_1!), we need to review the evidence the data provides against or in favour of H_0. This is done via a **test statistic** T that characterises the sample at hand. Agresti & Kateri (2022: 163) describe test statistics as indicators of how strongly a point estimate (e.g., a mean or a proportion) deviates from its expected value under H_0.

There are many possible test statistics out there:

- For instance, if the data are **discrete** and nominal, the χ^2 measure is used to compute differences between observed and expected frequencies in the entire sample.
- In the case of **continuous** data, it is common to rely on t for quantifying differences between sample means.
- Other possible test statistics include the correlation coefficient r, z-scores, the F-statistic, and many others.

Test statistics have characteristic probability distributions which are known as **sampling distributions** (e.g., the χ^2-distribution or the t-distribution). Recall that probability distributions are obtained by assigning probabilities to every outcome of a random variable.

Sampling distributions

A fundamental theoretical insight is that sample statistics (specifically means and sums of continuous random variables) converge onto a standard normal distribution with $\mu = 0$ and $\sigma = 1$. This proven statement is known as the **Central Limit Theorem** (Heumann, Schomaker, and Shalabh 2022: 547). However, if the population standard deviation is unknown, the Student t distribution provides a reasonable approximation. It has a single parameter v determining its shape, and it stands for the degrees of freedom.

t distribution

```
library(ggplot2)

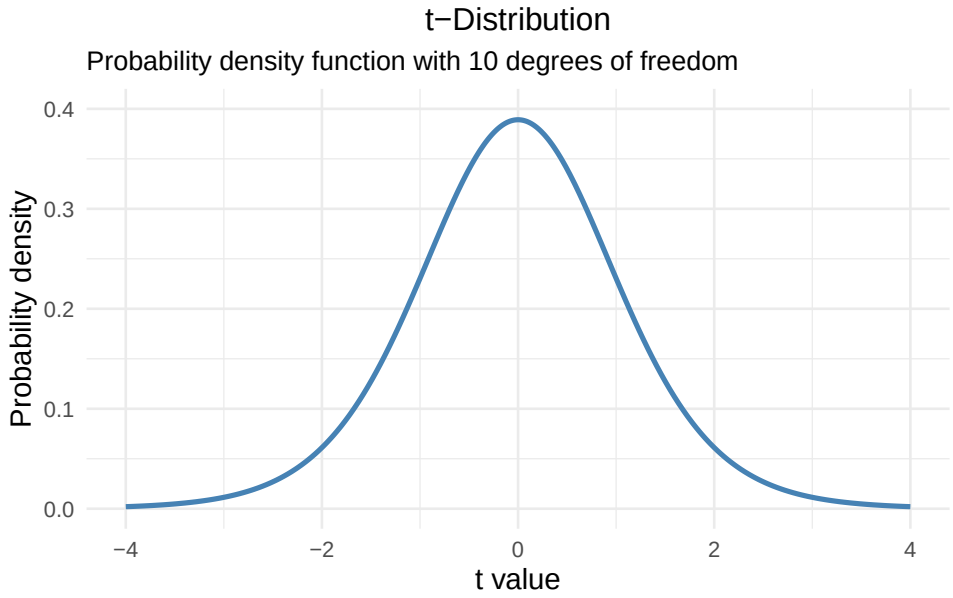
# Define the degrees of freedom
df_t <- 10

# Create a sequence of x values
x_t <- seq(-4, 4, length.out = 1000)

# Compute the t-distribution density
y_t <- dt(x_t, df = df_t)

# Create a data frame
t_distribution_data <- data.frame(x = x_t, y = y_t)

# Generate the plot
ggplot(t_distribution_data, aes(x = x, y = y)) +
  geom_line(color = "steelblue", linewidth = 1) + # Line for
   ↪  the density curve
  labs(
    title = "t-Distribution",
    subtitle = "Probability density function with 10 degrees
     ↪  of freedom",
    x = "t value",
    y = "Probability density"
  ) +
  theme_minimal() +
  coord_cartesian(ylim = c(0, 0.4), xlim = c(-4, 4)) +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.title = element_text(size = 12)
  )
```



The χ^2 distribution is closely related to the t distribution, sharing the degrees of freedom parameter v. In essence, it is the distribution of squared (standard) normally distributed random variables Z^2. Another example are sums of squares $\sum_{i=1}^{n}(x_i - \mu)^2$.

χ^2 distribution

```r
# Load ggplot2
library(ggplot2)

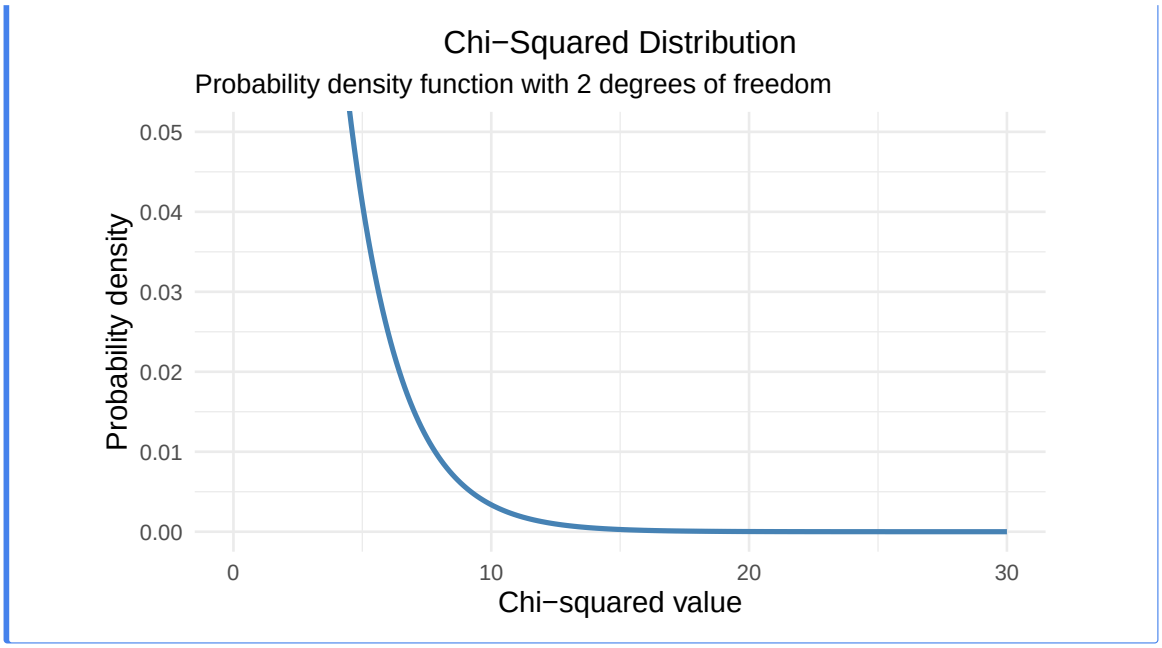
# Define the degrees of freedom
df <- 2

# Create a sequence of x values
x <- seq(0, 30, length.out = 1000)

# Compute the chi-squared density
y <- dchisq(x, df = df)

# Create a data frame
chi_squared_data <- data.frame(x = x, y = y)

# Generate the plot
ggplot(chi_squared_data, aes(x = x, y = y)) +
  geom_line(color = "steelblue", linewidth = 1) + # Line for
   ↪  the density curve
  labs(
    title = "Chi-Squared Distribution",
    subtitle = "Probability density function with 2 degrees
     ↪  of freedom",
    x = "Chi-squared value",
    y = "Probability density"
  ) +
  theme_minimal() +
  coord_cartesian(ylim = c(0, 0.05), xlim = c(0, 30)) +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.title = element_text(size = 12)
  )
```



Statistical significance

The final rejection of H_0 is determined by the **significance probability** p. Due to the frequency and ferocity with which statistical significance is misinterpreted in the research literature, we will begin by reviewing its technical definition:

> "The p-value is the probabilty, presuming that H_0 is true, that the test statistic equals the observed value or a value even more extreme in the direction predicted by H_a" (Agresti and Kateri 2022: 163).

In compact notation, it is equivalent to the conditional probability

$$P(T \geq \text{observed value} \mid H_0 \text{ is true}).$$

If p is lower than a pre-defined threshold (typically 0.05), also known as the **significance level** α, we can reject H_0. However, if $p \geq 0.05$, this **neither justifies rejecting nor accepting** the null hypothesis (Baguley 2012: 121).

What could go wrong? Type I and Type II errors

There is always a chance that we accept or reject the wrong hypothesis; the four possible constellations are summarised in the table below (cf. Heumann, Schomaker, and Shalabh 2022: 223):

	H_0 is true	H_0 is not true
H_0 is not rejected	Correct decision	Type II (β)-error
H_0 is rejected	Type I (α)-error	Correct decision

There is a trade-off between Type I and Type II errors (Agresti and Kateri 2022: 182-186):

- If we lower the significance level α, P(Type I) will decrease and P(Type II) will increase.
- Conversely, increasing α reduces β, but makes Type I errors more likely.
- Increasing the sample size is one way of reducing β (and increasing power) without chaning α.

i Statistical power

It is recommended to compute the **power** 1 - P(Type II) of a test, in order to estimate the probability of correctly rejecting H_0. The `pwr` library provides a convenient interface:

```
library(pwr)

# w = effect size (between 0 and 1)
# df = degrees of freedom
# sig.level = significance level alpha
# power = 1 - beta
pwr.chisq.test(w = 0.5, df = 1, sig.level = 0.05, power = 0.95)
```

```
     Chi squared power calculation

              w = 0.5
              N = 51.97884
             df = 1
      sig.level = 0.05
          power = 0.95

NOTE: N is the number of observations
```

```
# We'd need a sample size of N = 52 to detect an effect with effect
 ↪  size 0.5, alpha = 0.05, beta = 0.05, and 1 degree of freedom (low
 ↪  data complexity, e.g., simple 2x2 contingency table).
```

Where does the p-value come from?

Let's say that the statistical analysis of two discrete variables X and Y has returned a test statistic of $\chi^2 = 6.5$ for 2 df. In order to compute the corresponding p-value we need to consult the **sampling distribution** of this test statistic.

i χ^2 distribution

```r
# Load ggplot2
library(ggplot2)

# Define the degrees of freedom
df <- 2

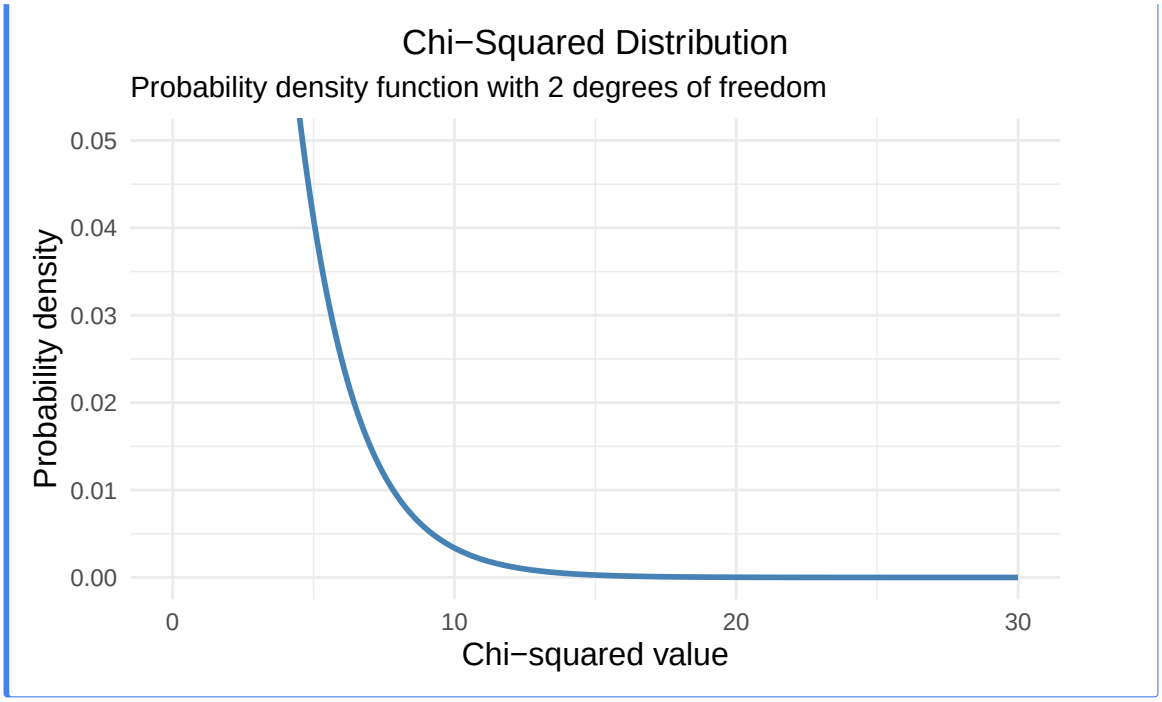
# Create a sequence of x values
x <- seq(0, 30, length.out = 1000)

# Compute the chi-squared density
y <- dchisq(x, df = df)

# Create a data frame
chi_squared_data <- data.frame(x = x, y = y)

# Generate the plot
ggplot(chi_squared_data, aes(x = x, y = y)) +
  geom_line(color = "steelblue", size = 1) +
  labs(
    title = "Chi-Squared Distribution",
    subtitle = "Probability density function with 2 degrees of
     ↪  freedom",
    x = "Chi-squared value",
    y = "Probability density"
  ) +
  theme_minimal() +
  coord_cartesian(ylim = c(0, 0.05), xlim = c(0, 30)) +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.title = element_text(size = 12)
  )
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```



Because sampling distributions are continuous probability distributions, they have a potentially infinite number of x-values. This also implies that the probability of any single x-value must be 0.[1]

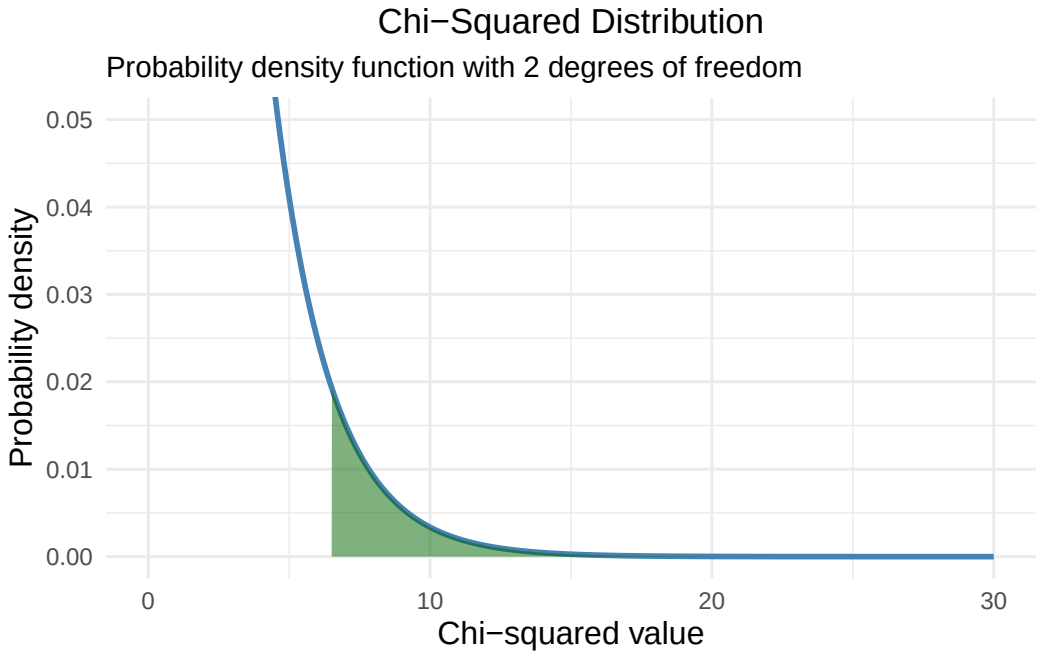
The sampling distribution thus has a probability density function (PDF), which we will call $f(x)$, and probabilities can only be obtained for an interval $[a, b]$. The probability that a value X falls into the interval $a < X < b$ is then equivalent to the area under the curve between a and b (cf. Equation 8).

$$P(a < X < b) = \int_a^b f(x)dx. \tag{8}$$

Recall the PDF $f_{\chi^2}(x)$ of the χ^2-distribution with 2 degrees of freedom. The p-value corresponds to the green area under the curve ranging from $x = 6.5$ up to ∞, which can be restated formally in Equation 9. This brings us back to the definition of the p-value: It is the probability that the χ^2 score is equal to 6.5 or **higher**, i.e., $P(\chi^2 \geq 6.5)$.

$$P(\chi^2 \geq 6.5) = P(6.5 < \chi^2 < \infty) = \int_{6.5}^{\infty} f_{\chi^2}(x)dx. \tag{9}$$

[1]The proof is given in Heumann et al. (2022: 544).



In practice, statistical software computes this integral for us:

```r
pchisq(6.5, df = 2, lower.tail = FALSE)
```

```
[1] 0.03877421
```

Practical considerations

! Common pitfalls [cf. @agrestiFoundationsStatisticsData2022: 189-190]

- Statistical significance is **NOT** an indication of a causal relationship between the variables of interest (**correlation** $\neq$ **causation**).
- p-values do **NOT** signify the strength of an effect ($\neq$ effect size). It only helps identify whether there is an effect to begin with.
- p-values are **NOT** the probability of the null hypothesis being true.
- Statistical significance is only a starting point for further scientific inquiry, and by no means the end of it.

Exercises

Tier 1

Exercise 1. A hypothesis test has returned a p-value of 0.03. Which of the following interpretations is correct? Explain briefly why the other options are incorrect.

a. The probability that H_0 is true is 0.03.
b. Assuming H_0 is true, the probability of observing a result at least as extreme as the observed one is 0.03.
c. There is a 97% chance that the alternative hypothesis is true.

Exercise 2. A study reports:

> The result was not statistically significant.

- Can we conclude there is no effect?
- Can we conclude that the null hypothesis is true?
- What *can* we conclude?

Tier 2

Exercise 3. For each of the research questions below, formulate a H_0 and a H_1:

a. Does the use of progressive aspect depend on text genre?
b. Do native and learner speakers differ in their average sentence length?
c. Is word frequency correlated with age of acquisition?

Exercise 4. A researcher plans a study and fails to properly control for statistical power, limiting it to 0.01. Interpret what a power of 0.01 means in terms of long-run decision outcomes when H_1 is true. Compare it to a study with a power of 0.80 – how does the reliability of conclusions change?

Tier 3

Exercise 5. Suppose a χ^2-test produces a test statistic $\chi^2 = 8.2$ for degrees of freedom $df = 3$. Compute the corresponding p-value and verify that the result corresponds to the shaded tail area under the χ^2-distribution. Explain why the integral begins at 8.2 and not at 0.

> **Hint**
>
> Adjust the sample code from the last section (degrees of freedom and observed test statistic) to match the values given in the task:

```r
# Define the degrees of freedom
df <- 2

# Create a sequence of x values
x <- seq(0, 30, length.out = 1000)

# Compute the chi-squared density
y <- dchisq(x, df = df)

# Create a data frame
chi_squared_data <- data.frame(x = x, y = y)

# Define the integral for highlighting
highlight_data <- subset(chi_squared_data, x >= 6.5 & x <= 20)

# Generate the plot
ggplot(chi_squared_data, aes(x = x, y = y)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_area(data = highlight_data, aes(x = x, y = y),
            fill = "darkgreen", alpha = 0.5) + # integral
  labs(
    title = "Chi-Squared Distribution",
    subtitle = "Probability density function with 2 degrees of
    ↪  freedom",
    x = "Chi-squared value",
    y = "Probability density"
  ) +
  theme_minimal() +
  coord_cartesian(ylim = c(0, 0.05), xlim = c(0, 30)) +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.title = element_text(size = 12)
  )
```

References

Agresti, Alan, and Maria Kateri. 2022. *Foundations of Statistics for Data Scientists: With r and Python*. Boca Raton: CRC Press.

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